Description:
Developed a music-reactive LED strip system using Arduino Nano and a sound sensor to create dynamic lighting effects based on audio input. The circuit uses an IRFZ44N MOSFET and BC547 transistor for efficient switching and amplification, enabling real-time LED control synchronized with music beats. The project demonstrates skills in embedded systems, circuit design, and hardware interfacing.

Music Beat LED Strip (Arduino Nano) – Parts List

• Arduino Nano
• Sound Sensor Module
• PCB Board (General Purpose)
• IRFZ44N MOSFET (for LED strip control)
• BC547 NPN Transistor
• 10kΩ Resistors (2x)
• Screw Terminals (2x)
• Female Headers (for connections)

Extra (optional but useful):
• 12V LED Strip
• 12V Power Supply
• Jumper Wires

🔌 **1. Sound Sensor → Arduino Nano**

- VCC → 5V
- GND → GND
- DO → D2 (Digital Pin)
  (AO use nahi karna ho to skip kar sakte ho)

⚡ **2. MOSFET (IRFZ44N) Connection (LED Strip Control)**

IRFZ44N pins (front side se): **Gate – Drain – Source**

- **Gate → Arduino D9 (PWM pin)**
- **Source → GND**
- **Drain → LED Strip Negative (-)**

💡 **3. LED Strip + Power**

- LED Strip **Positive (+) → 12V Power Supply**
- LED Strip **Negative (–) → MOSFET Drain**

🔧 **4. BC547 Transistor (Signal Amplification)**

- **Base → Sound Sensor DO (via 10k resistor)**
- **Collector → Arduino D2**
- **Emitter → GND**

🔩 **5. Resistors**

- 10k between **Base (BC547) & Sensor DO**
- 10k between **Gate (MOSFET) & GND** *(pull-down, important)*

🔌 **6. Common Ground (VERY IMPORTANT)**

👉 Arduino GND + Sensor GND + Power Supply GND → **Sab ek sath connect karo**

⚠️ **Important Tips**

- LED strip ke liye **external 12V supply use karo**
- Arduino se direct LED strip mat chalana ❌
- MOSFET heat ho to heat sink lagao